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Nested Cross-Validated Double Machine Learning for the Partially Linear IV Model

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Summary Double/debiased machine learning (DML) enables $\sqrt{n}$ inference in partially linear IV models when nuisance regressions are estimated flexibly, but inference can be sensitive to how nuisance learners are chosen and tuned. This paper studies a leakage-safe nested cross-validation protocol inside outer cross-fitting folds: nuisance learners are selected and tuned using training data only, refit on the full training fold, and evaluated on held-out observations in the residualized PLIV score. For finite searched libraries, I derive a benchmark-relative excess-risk bound under a balanced foldwise margin and conditional Bernstein-type fast-rate condition, yielding a sufficient condition for the usual PLIV DML product-rate requirement. Under strong identification, the resulting estimator is asymptotically normal. Simulations and a CollegeDistance application show competitive performance and reduced sensitivity to arbitrary fixed learner choices.

Keywords: Double/debiased machine learning, instrumental variables, nested cross-validation, unobserved confounding, high-dimensional econometrics.
JEL Codes: C14, C21, C26, C55

1. INTRODUCTION

Instrumental variables (IV) methods are a central tool for causal inference when treatment assignment is endogenous. By leveraging exogenous variation induced by a valid instrument, IV can identify causal effects even in the presence of unobserved confounding Angrist and Krueger (1991); Angrist and Imbens (1994); Heckman and Vytlacil (2005). Classical econometric theory provides a rich set of estimators for linear and nonlinear IV models, including nonlinear two-stage least squares Amemiya (1974) and efficient IV estimation under conditional moment restrictions Chamberlain (1987); Newey (1990). In modern applications, however, researchers often face two practical complications simultaneously: (i) *high-dimensional* controls and rich instrument sets, and (ii) a desire to allow for flexible nonlinear reduced-form relationships in the first stage and/or control function. In such settings, naive extensions of parametric IV can be sensitive.

A first challenge is the interaction of high dimensionality and instrument strength. With many controls or instruments, standard asymptotics can be unreliable and finite-sample bias can be severe when instruments are weak Staiger and Stock (1997); Andrews and Stock (2005). A parallel literature studies many instrument environments and regularization strategies Hansen, Hausman, and Newey (2008); Carrasco (2012); Hansen and Kozbur (2014); Bai and Ng (2009), and develops high-dimensional IV estimators and inference methods under sparsity or structural restrictions Caner (2009); Gautier and Tsybakov (2011); Gold, Lederer, and Tao (2020); Belloni et al. (2012). A second challenge is functional-form uncertainty: relationships such as $E[D \mid Z, W]$ and $E[Y \mid W]$ may be nonlinear and high-dimensional. Machine learning (ML) provides flexible tools for prediction and can mitigate misspecification in rich environments Mullainathan and Spiess (2017); Athey and Imbens (2019). In IV contexts, ML has been used to estimate reduced

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forms and even nonlinear structural relationships (DeepIV) Hartford et al. (2017), and recent work studies deep-learning-based estimation and inference in econometrics more broadly Farrell et al. (2021); Chen et al. (2022).

A central obstacle to combining ML with IV is that regularization and model selection can induce non-negligible bias in plug-in estimators of causal parameters. Double/debiased machine learning (DML) resolves this problem by combining (i) *orthogonal* (locally robust) score functions—in the sense of Neyman orthogonality Neyman (1959, 1979)—and (ii) cross-fitting Chernozhukov et al. (2018); Newey and Robins (2018). Orthogonality makes the identifying moment locally insensitive to nuisance estimation error, while cross-fitting prevents overfitting bias from contaminating the score evaluation. Locally robust semiparametric estimation provides a complementary lens on these constructions Chernozhukov et al. (2022a). The DML template has been adapted to a wide range of causal and structural settings, including difference-in-differences Chang (2020), logistic partially linear models Liu, Zhang and Zhou (2021), and dynamic treatment effects Bodory, Huber and Lafférs (2022). Knaus (2022) provides applied guidance emphasizing the practical steps required for reliable DML implementation.

While DML theory is explicit about the properties required of nuisance estimators, applied implementations still require the researcher to choose a nuisance learning algorithm and its tuning parameters. This choice is consequential: the nuisance regressions can be high-dimensional, nonlinear, and noisy, and a fixed default (such as always using Lasso or Random Forest) can perform poorly when the true nuisance functions are not well approximated by the chosen model class. Moreover, researcher degrees of freedom in nuisance selection can threaten reproducibility. In practice, researchers often choose among penalized linear learners Hoerl and Kennard (1970); Tibshirani (1996); Zou and Hastie (2005), ensemble methods such as Random Forests and boosting Breiman (2001); Friedman (2001), and neural networks Goodfellow et al. (2016); see also the general ML treatment in Hastie et al. (2009). The question is how to *select and tune* among these candidates without violating the sample-splitting logic that delivers valid DML inference.

This paper studies leakage-safe fold-specific nuisance tuning for DML in the partially linear instrumental variables (PLIV) model $Y = \tau_0 D + g_0(W) + \zeta,\ E[\zeta \mid Z, W] = 0$, where W may be high-dimensional. I work with the standard residualized orthogonal score underlying residualized 2SLS/GMM, $\psi(\tau, \eta) = (Z - \pi(W))\big[(Y - \mu(W)) - \tau(D - r(W))\big],\ \eta = (\pi, \mu, r)$, where $\pi_0(W) = E[Z \mid W]$, $\mu_0(W) = E[Y \mid W]$, and $r_0(W) = E[D \mid W]$. The econometric issue is not the score itself but nuisance selection: applied implementations must choose learners and hyperparameters, often by full-sample cross-validation or by tuning once and reusing the result across folds, which can leak evaluation observations into nuisance fitting and blur the train/evaluate separation required by cross-fitting. I therefore propose a nested cross-validation protocol within each outer fold: learners and tuning parameters are selected by inner cross-validation using only the outer-fold training sample, refit on that training sample, and evaluated only on the held-out fold. Under standard strong-identification and high-level product-rate conditions for the selected nuisances, the resulting estimator retains the familiar orthogonality-based remainder bounds and the usual asymptotic normality and plug-in variance formula of cross-fitted PLIV DML.

The contribution is threefold. First, the paper formalizes a leakage-safe protocol for fold-specific nuisance tuning in cross-fitted PLIV estimation. Second, for finite searched libraries under squared-loss inner cross-validation, it derives a selector-specific excess-risk bound under a balanced foldwise margin and conditional Bernstein-type fast-rate condition: the selected refitted nuisance achieves benchmark-relative excess risk within the searched library up to a $\log N/n$ term and a refit-stability remainder, yielding a finite-library sufficient condition for the standard PLIV DML product-rate requirement. Third, Monte Carlo evidence and the `CollegeDistance` application show that disciplined nested selection can reduce sensitivity to arbitrary fixed learner choices while preserving competitive finite-sample performance.

This paper differs from NPIV work that targets a structural function $h(D, W)$ under completeness-type conditions Newey and Powell (2003); Darolles et al. (2011); Bennett et al. (2022); Chen and Pouzo (2012); Horowitz and Lee (2007); Horowitz (2011); from work that learns optimal instruments or automatic debiasing weights Chen et al. (2021); Chernozhukov, Newey, and Singh (2022b,c); from Çetin (2026), which studies inference-targeted selection of the debiasing/Riesz component for DML estimators of linear functionals; and from work on invalid instruments and selective ML choices Sun, Cui, and Tchetgen (2022). Here I take the standard residualized PLIV score as given and ask a narrower question: when nuisance learners are selected separately within each outer training fold, can that finite-library search rule be linked formally to the nuisance-rate conditions used in cross-fitted PLIV DML, and how much practical sensitivity can such a protocol discipline?

The remainder of the paper is structured as follows. Section 2 presents the PLIV model, the orthogonal score, and the nested-CV cross-fitted estimator. Section 3 gives identification, orthogonality, a selector-specific nuisance-rate transfer result for the finite searched-library setting, and asymptotic normality. Section 4 reports Monte Carlo evidence, and Section 5 presents the empirical application. Section 6 concludes. Additional proofs and supplementary results appear in the Online Appendix.

2. MODEL AND IDENTIFICATION

2.1. Setup: partially linear IV with high-dimensional controls

Let $\{(Y_i, D_i, Z_i, W_i)\}_{i=1}^n$ be an i.i.d. sample, where $Y_i \in \mathbb{R}$ is the outcome, $D_i \in \mathbb{R}$ is an endogenous regressor (treatment), $W_i \in \mathbb{R}^p$ is a potentially high-dimensional vector of observed controls, and $Z_i \in \mathbb{R}^{d_z}$ is a (possibly vector-valued) instrument with fixed dimension $d_z \geq 1$.[1]

The partially linear IV (PLIV) model is

$$Y_i = \tau_0 D_i + g_0(W_i) + \zeta_i, \qquad E[\zeta_i \mid Z_i, W_i] = 0, \tag{2.1}$$

where τ_0 is the scalar parameter of interest and $g_0(\cdot)$ is an unknown regression function that collects the direct effect of W on Y.

Assumption 1 (Instrument exogeneity)**.** *The structural error satisfies $E[\zeta_i \mid Z_i, W_i] = 0$ in (2.1).*

Assumption 2 (Instrument relevance / rank condition)**.** *Let $\tilde{Z}_i := Z_i - \pi_0(W_i)$ and $\tilde{D}_i := D_i - r_0(W_i)$, where $\pi_0(W) = E[Z \mid W]$ and $r_0(W) = E[D \mid W]$. Assume that $\Omega_0 := E[\tilde{Z}_i \tilde{Z}_i']$ is positive definite and $g_{ZD,0} := E[\tilde{Z}_i \tilde{D}_i] \neq 0$.*

Assumption 2 is the usual IV relevance condition after partialling out W; it ensures that the residualized instrument $\tilde{Z}$ carries identifying variation for the residualized endogenous regressor $\tilde{D}$.

2.2. Orthogonalization and identifying moments

Define the nuisance regressions

$$\mu_0(W) := E[Y \mid W], \qquad r_0(W) := E[D \mid W], \qquad \pi_0(W) := E[Z \mid W],$$

[1] The scalar-instrument case corresponds to $d_z = 1$ and is a special case of the analysis below.

and the residualized variables

$$\tilde{Y}_i := Y_i - \mu_0(W_i), \qquad \tilde{D}_i := D_i - r_0(W_i), \qquad \tilde{Z}_i := Z_i - \pi_0(W_i).$$

This orthogonalization strategy builds directly on the foundational partialling-out approach of Robinson (1988) for partially linear models, extending it to the instrumental variables setting to isolate the exogenous variation. The regression $\pi_0(W) = E[Z|W]$ is not a structural model for the instrument; it is the projection used to residualize Z with respect to W, exactly as in classical IV with controls.

Using (2.1) and iterated expectations, one can write

$$\tilde{Y}_i = \tau_0 \tilde{D}_i + \zeta_i, \qquad E[\zeta_i \mid Z_i, W_i] = 0,$$

so that the identifying restriction becomes the vector moment condition

$$E\Big[\tilde{Z}_i\big(\tilde{Y}_i - \tau_0 \tilde{D}_i\big)\Big] = 0 \in \mathbb{R}^{d_z}. \tag{2.2}$$

This moment condition is equivalent to the population normal equations of IV on residuals, i.e. 2SLS after partialling out W. Let $\eta = (\pi, \mu, r)$ denote candidate nuisance functions. Define the (vector) Neyman-orthogonal score

$$\psi_i(\tau, \eta) = \big(Z_i - \pi(W_i)\big)\Big[\big(Y_i - \mu(W_i)\big) - \tau\,\big(D_i - r(W_i)\big)\Big] \in \mathbb{R}^{d_z}. \tag{2.3}$$

Under Assumptions 1–2, τ_0 solves $E[\psi_i(\tau, \eta_0)] = 0$, where $\eta_0 = (\pi_0, \mu_0, r_0)$.

2.3. Connection to residualized 2SLS (oracle benchmark)

If (μ_0, r_0, π_0) were known, the IV estimator based on (2.2) is the residualized 2SLS coefficient:

$$\tau_0 = \arg\min_{\tau \in \mathbb{R}} \Big\| E[\tilde{Z}_i(\tilde{Y}_i - \tau \tilde{D}_i)] \Big\|^2_{\Omega_0^{-1}} \quad \Longrightarrow \quad \tau_0 = \frac{g'_{ZD,0}\Omega_0^{-1} g_{ZY,0}}{g'_{ZD,0}\Omega_0^{-1} g_{ZD,0}},$$

where $g_{ZY,0} := E[\tilde{Z}_i \tilde{Y}_i]$ and $\|a\|^2_{\Omega^{-1}} := a'\Omega^{-1}a$. When $d_z = 1$, this reduces to the familiar IV ratio. The DML estimator below replaces (μ_0, r_0, π_0) with cross-fitted ML estimates while maintaining the same residualized 2SLS/GMM structure[2].

2.4. Estimation: cross-fitting with nested cross-validation

The estimator combines outer cross-fitting with inner-fold learner selection and tuning restricted to the outer-fold training sample. In each outer fold, learners for $\mu_0(W) = E[Y \mid W]$, $r_0(W) = E[D \mid W]$, and $\pi_0(W) = E[Z \mid W]$ are chosen by inner cross-validation on the training data, refit on the full training fold, and evaluated only on the held-out fold. The resulting cross-fitted residuals are then aggregated into the residualized 2SLS/GMM estimator and its plug-in variance formula. For $d_z > 1$, the instrument regression is implemented coordinate-wise. The full step-by-step algorithm is reported in the Online Appendix.

[2]To avoid ambiguity, subscripts "0" denote oracle (population) nuisance functions and their residuals. Define the oracle residual random variables $\tilde{Y}_0 := Y - \mu_0(W)$, $\tilde{D}_0 := D - r_0(W)$, and $\tilde{Z}_0 := Z - \pi_0(W)$, with observation-level versions $\tilde{Y}_{0,i} := Y_i - \mu_0(W_i)$, $\tilde{D}_{0,i} := D_i - r_0(W_i)$, and $\tilde{Z}_{0,i} := Z_i - \pi_0(W_i)$. The cross-fitted residuals used by the estimator are $\tilde{Y}_i := Y_i - \hat{\mu}^{(-k(i))}(W_i)$, $\tilde{D}_i := D_i - \hat{r}^{(-k(i))}(W_i)$, and $\tilde{Z}_i := Z_i - \hat{\pi}^{(-k(i))}(W_i)$, where $k(i)$ denotes the outer fold containing observation i.

3. MAIN RESULTS

Assumption 3 (Regularity for differentiation and inference)**.** *The numbers of outer and inner folds, K and J, are fixed, and the sample partitions are asymptotically balanced, so*

$$|\mathcal{I}_k|/n \to 1/K, \qquad |\mathcal{I}_{-k,m}|/n \to (K-1)/(KJ).$$

For each admissible direction $h = (h_\pi, h_\mu, h_r)$ used in Lemma 1, one has

$$E\big[\|h_\pi(W)\|_2^2 + h_\mu(W)^2 + h_r(W)^2\big] < \infty,$$

the path $\eta_0 + \varepsilon h$ belongs to the nuisance space for all sufficiently small $|\varepsilon|$, and

$$\sup_{|\varepsilon|\le\varepsilon_0} \|\psi(O;\tau_0,\eta_0+\varepsilon h)\|_2$$

is dominated by an integrable envelope. Moreover, there exists a finite constant C such that

$$E[\zeta^2 \mid W] \le C, \qquad E[\|Z-\pi_0(W)\|_2^2 \mid W] \le C, \qquad E[(D-r_0(W))^2 \mid W] \le C$$

almost surely, and for some $\delta > 0$,

$$E[\|\psi(O;\tau_0,\eta_0)\|_2^{2+\delta}] < \infty.$$

Finally,

$$\sup_{1\le k\le K}\Big(\|\hat\mu^{(-k)}\|_\infty + \|\hat r^{(-k)}\|_\infty + \|\hat\pi^{(-k)}\|_\infty\Big) = O_p(1),$$

and

$$\|\mu_0\|_\infty + \|r_0\|_\infty + \|\pi_0\|_\infty < \infty.$$

3.1. Identification

Theorem 1 (Identification). Under Assumptions 1–2, the parameter τ_0 is uniquely identified by the vector moment condition $E[\psi_i(\tau,\eta_0)] = 0$, where ψ_i is defined in (2.3).

Proof. See Online Appendix B.1.

3.2. Neyman orthogonality

Define the population moment mapping $M(\tau,\eta) := E[\psi_i(\tau,\eta)] \in \mathbb{R}^{d_z}$. Neyman orthogonality means that small perturbations of the nuisance functions do not affect $M(\tau_0,\eta)$ at first order.

Lemma 1 (First-order (Gateaux) orthogonality). Let $\psi_i(\tau,\eta)$ be defined in (2.3). For any admissible square-integrable direction $h = (h_\pi, h_\mu, h_r)$ satisfying Assumption 3, define the Gateaux derivative

$$D_\eta M(\tau,\eta)[h] := \frac{d}{d\varepsilon} M(\tau,\eta+\varepsilon h)\Big|_{\varepsilon=0}.$$

Under Assumptions 1 and 3, with $\eta_0 = (\pi_0,\mu_0,r_0)$ and $\pi_0(W) = E[Z \mid W]$, $\mu_0(W) = E[Y \mid W]$, $r_0(W) = E[D \mid W]$,

$$D_\eta M(\tau_0,\eta_0)[h] = 0 \in \mathbb{R}^{d_z} \qquad \text{for every such admissible direction } h.$$

Proof. See Online Appendix B.2.

3.3. Second-order structure and double robustness

The orthogonal PLIV score has a convenient *exact* decomposition: the population moment at τ_0 equals a product of nuisance errors. This fact both (i) clarifies why first-order terms vanish and (ii) yields sharp second-order remainder bounds.

Lemma 2 (Second-order moment decomposition). Let $m(W) := \mu(W) - \tau_0 r(W)$ and $m_0(W) := \mu_0(W) - \tau_0 r_0(W) = E[Y - \tau_0 D \mid W]$. For any candidate $\eta = (\pi, \mu, r)$,

$$M(\tau_0, \eta) = E[(\pi_0(W) - \pi(W))\,(m_0(W) - m(W))] \in \mathbb{R}^{d_z}. \tag{3.4}$$

In particular, $\|M(\tau_0, \eta)\|_2 \le \|\pi - \pi_0\|_{P,2} \Big(\|\mu - \mu_0\|_{P,2} + |\tau_0|\, \|r - r_0\|_{P,2} \Big)$, and $M(\tau_0, \eta) = 0$ whenever either (i) $\pi = \pi_0$ a.s. or (ii) $\mu = \mu_0$ and $r = r_0$ a.s.

Proof. See Online Appendix B.3.

Lemma 2 is the formal "second-order orthogonality" statement: the bias induced by nuisance misspecification enters as a product of nuisance errors. The usual $n^{-1/4}$ rate requirements are exactly what ensure this product is $o_p(n^{-1/2})$.

Lemma 3 (Foldwise second-order bound under nested-CV cross-fitting). Let $\hat{\eta}^{(-k)} = (\hat{\pi}^{(-k)}, \hat{\mu}^{(-k)}, \hat{r}^{(-k)})$ be nuisance estimators constructed using only the training sample $\mathcal{I}_{-k}$ in outer fold k (including inner CV selection/tuning). Then, conditional on the training sample $\mathcal{I}_{-k}$,

$$\Big\| E\Big[\psi_i(\tau_0, \hat{\eta}^{(-k)}) - \psi_i(\tau_0, \eta_0) \mid \mathcal{I}_{-k}\Big] \Big\|_2 \lesssim \|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} \Big(\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + |\tau_0|\, \|\hat{r}^{(-k)} - r_0\|_{P,2} \Big).$$

Consequently, if $\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} = o_p(n^{-1/4})$ and $\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + \|\hat{r}^{(-k)} - r_0\|_{P,2} = o_p(n^{-1/4})$ uniformly in k, then the foldwise bias term is $o_p(n^{-1/2})$.

Proof. See Online Appendix B.4.

Remark 1 (Nested-CV selection and cross-fitting measurability). The nested-CV protocol selects learners and tuning parameters using only the training sample $\mathcal{I}_{-k}$ within each outer fold k. Hence the fitted nuisances $\hat{\eta}^{(-k)} = (\hat{\mu}^{(-k)}, \hat{r}^{(-k)}, \hat{\pi}^{(-k)})$ are measurable with respect to $\sigma(\{(Y_i, D_i, Z_i, W_i) : i \in \mathcal{I}_{-k}\})$ and do not depend on the held-out fold $\mathcal{I}_k$. Conditional on $\mathcal{I}_{-k}$, the score evaluations on $\mathcal{I}_k$ are therefore computed on an independent subsample, so the standard cross-fitting orthogonality expansion applies. The rate conditions in Theorem 2 are imposed on these *selected* fold-specific nuisances. Section 3.4 provides a selector-specific sufficient condition linking the nested-CV rule to those rates in the finite searched-library setting used.

Remark 2 (Scope of the theoretical result). The paper does not derive a selector-specific semiparametric efficiency theory for arbitrary data-adaptive learner search. Instead, the new result below works in the finite searched-library, squared-loss setting used in the simulations and empirical implementation. Under a balanced foldwise margin and conditional Bernstein-type fast-rate condition, the selected refitted nuisance achieves benchmark-relative excess risk up to a $\log N/n$ term and a refit-stability remainder. For fixed searched libraries and sufficiently small refit-stability errors, this transfers the benchmark $L_2(P)$ rate to the selected nuisance and closes the bridge to the PLIV orthogonality expansion. These are high-level sufficient conditions and are not verified separately for each off-the-shelf learner used in Sections 4–5; the empirical library should therefore be read as illustrative of modern applied practice rather than as a primitive learner-by-learner theory exercise.

3.4. Nested-CV selection and nuisance-rate transfer

To connect fold-specific nested-CV selection to the high-level nuisance-rate conditions, I work in the finite searched-library setting used in the simulations and empirical implementation. I formulate the result for a generic scalar nuisance target T with regressors X, covering $T = Y$, $T = D$, and each coordinate $T = Z_j$. For an outer fold k, let $\mathcal{G}_{T,n}$ denote the searched library of candidate learner for T. For each $g \in \mathcal{G}_{T,n}$, let $\hat{f}_g^{(-k,-m)}$ denote the fit obtained on the m-th inner training split, and define the inner-CV score

$$\widehat{R}_{T,k}(g) = \frac{1}{J}\sum_{m=1}^{J} \frac{1}{|\mathcal{I}_{-k,m}|} \sum_{i \in \mathcal{I}_{-k,m}} \big(T_i - \hat{f}_g^{(-k,-m)}(X_i)\big)^2.$$

Let $\hat{g}_{T,k} \in \arg\min_{g \in \mathcal{G}_{T,n}} \widehat{R}_{T,k}(g)$, with deterministic tie-breaking, and let $\hat{f}_{\hat{g}_{T,k}}^{(-k)}$ denote the refit of the selected learner on the full outer training sample $\mathcal{I}_{-k}$.

Assumption 4 (Finite searched library)**.** *For each nuisance target $T \in \{Y, D, Z_1, \ldots, Z_{d_z}\}$ and each outer fold k, nested cross-validation searches over a finite library $\mathcal{G}_{T,n}$ with cardinality $N_{T,n} = |\mathcal{G}_{T,n}|$.*

Define the inner-training sigma-field for outer fold k and inner fold m by

$$\mathcal{F}_{k,m}^{\mathrm{tr}} := \sigma\{O_i : i \in \mathcal{I}_{-k} \setminus \mathcal{I}_{-k,m}\}.$$

Define the average inner-risk benchmark

$$\bar{R}_{T,k}(g) := \frac{1}{J}\sum_{m=1}^{J} E\Big[\big(T - \hat{f}_g^{(-k,-m)}(X)\big)^2 \,\Big|\, \mathcal{F}_{k,m}^{\mathrm{tr}}\Big].$$

Let $g_{T,n}^{\dagger} \in \mathcal{G}_{T,n}$ denote a nonrandom searched-library benchmark candidate. For each $g \in \mathcal{G}_{T,n}$, define the benchmark-relative gap

$$\bar{\Gamma}_{T,k}(g) := \bar{R}_{T,k}(g) - \bar{R}_{T,k}(g_{T,n}^{\dagger}),$$

and, for each inner fold m,

$$\Delta_{g,k,m}(O) := \big(T - \hat{f}_g^{(-k,-m)}(X)\big)^2 - \big(T - \hat{f}_{g_{T,n}^{\dagger}}^{(-k,-m)}(X)\big)^2,$$

$$\gamma_{g,k,m} := E\big[\Delta_{g,k,m}(O) \,\big|\, \mathcal{F}_{k,m}^{\mathrm{tr}}\big].$$

Assumption 5 (Benchmark quality and refit stability)**.** *For each T, there exist deterministic sequences $b_{T,n} \downarrow 0$ and $s_{T,n} \downarrow 0$ and a nonrandom benchmark candidate $g_{T,n}^{\dagger} \in \mathcal{G}_{T,n}$ such that, uniformly in k:*

(i) Benchmark quality:

$$R_{T,k}^{\mathrm{full}}(g_{T,n}^{\dagger}) - R_{T,0} = O_p(b_{T,n}^2),$$

where

$$R_{T,k}^{\mathrm{full}}(g) := E\Big[\big(T - \hat{f}_g^{(-k)}(X)\big)^2 \,\Big|\, \mathcal{I}_{-k}\Big], \qquad R_{T,0} := E\big[(T - m_{T,0}(X))^2\big],$$

and $m_{T,0}(X) = E[T \mid X]$.

(ii) Refit stability:

$$\sup_{g \in \mathcal{G}_{T,n}} \left| R^{\text{full}}_{T,k}(g) - \bar{R}_{T,k}(g) \right| = O_p(s_{T,n}).$$

Assumption 6 (Balanced foldwise margin and conditional Bernstein-type fast-rate condition)**.** *There exist constants $\underline{c} \in (0,1]$ and $C, B \in (0,\infty)$ such that, uniformly over outer folds k, all candidates $g \in \mathcal{G}_{T,n}$ satisfying $\bar{\Gamma}_{T,k}(g) > 0$, and all inner folds $m \in \{1,\dots,J\}$:*

(i) Balanced foldwise margin:

$$\gamma_{g,k,m} \geq \underline{c}\, \bar{\Gamma}_{T,k}(g).$$

(ii) Bernstein variance control:

$$\operatorname{Var}\big(\Delta_{g,k,m}(O) \,\big|\, \mathcal{F}^{\text{tr}}_{k,m}\big) \leq C\, \gamma_{g,k,m}.$$

(iii) Conditional exponential moment:

$$E\left[\exp\left(\frac{|\Delta_{g,k,m}(O) - \gamma_{g,k,m}|}{B}\right) \,\middle|\, \mathcal{F}^{\text{tr}}_{k,m}\right] \leq 2.$$

Lemma 4 (Fast-rate benchmark-relative excess-risk bound)*.* Suppose Assumptions 4–6 hold for T. Then

$$\left(R^{\text{full}}_{T,k}(\hat{g}_{T,k}) - R^{\text{full}}_{T,k}(g^{\dagger}_{T,n})\right)_+ = O_p\left(\frac{\log N_{T,n}}{n} + s_{T,n}\right),$$

uniformly in k.

Lemma 5 (Rate transfer from benchmark to selected refit)*.* Suppose Lemma 4 and Assumption 5 hold for a nuisance target T. Then

$$R^{\text{full}}_{T,k}(\hat{g}_{T,k}) - R_{T,0} = O_p\left(b^2_{T,n} + \frac{\log N_{T,n}}{n} + s_{T,n}\right),$$

uniformly in k. Consequently,

$$\|\hat{f}^{(-k)}_{\hat{g}_{T,k}} - m_{T,0}\|_{P,2} = O_p(\rho_{T,n}), \qquad \rho_{T,n} := b_{T,n} + \sqrt{\frac{\log N_{T,n}}{n}} + \sqrt{s_{T,n}},$$

uniformly in k.

Corollary 1 (Finite-library sufficient condition for Theorem 2)*.* Apply Lemma 5 to $T = Y$, $T = D$, and each $T = Z_j$, and define

$$\rho_{\mu,n} := b_{\mu,n} + \sqrt{\frac{\log N_{\mu,n}}{n}} + \sqrt{s_{\mu,n}}, \qquad \rho_{r,n} := b_{r,n} + \sqrt{\frac{\log N_{r,n}}{n}} + \sqrt{s_{r,n}},$$

and

$$\rho_{\pi,n} := \max_{1 \leq j \leq d_z} \left(b_{\pi_j,n} + \sqrt{\frac{\log N_{\pi_j,n}}{n}} + \sqrt{s_{\pi_j,n}}\right).$$

Because d_z is fixed,

$$\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} = O_p(\rho_{\pi,n})$$

uniformly in k. If

$$\rho_{\mu,n} = o(1), \qquad \rho_{r,n} = o(1), \qquad \rho_{\pi,n} = o(1),$$

and

$$\rho_{\pi,n}\big(\rho_{\mu,n} + |\tau_0|\,\rho_{r,n}\big) = o(n^{-1/2}),$$

then the selected fold-specific nuisances satisfy the assumptions of Theorem 2. In particular, if each searched library has fixed cardinality, $s_{\mu,n} = o(n^{-1/2})$, $s_{r,n} = o(n^{-1/2})$, and $\max_j s_{\pi_j,n} = o(n^{-1/2})$, and the benchmark candidates satisfy

$$b_{\mu,n} = o(n^{-1/4}), \qquad b_{r,n} = o(n^{-1/4}), \qquad \max_{1\le j\le d_z} b_{\pi_j,n} = o(n^{-1/4}),$$

then Theorem 2 follows.

3.5. Asymptotic normality and inference

I state the standard DML result for the nested-CV cross-fitted residualized 2SLS estimator. The key inputs are: (i) Neyman orthogonality (Lemma 1), (ii) second-order remainder control (Lemmas 2–3), and (iii) a central limit theorem for the orthogonal score.

Theorem 2 (Consistency and asymptotic normality). Suppose Assumptions 1–2 and 3 hold and the cross-fitted nuisance estimators satisfy, uniformly over folds k,

$$\|\hat\mu^{(-k)} - \mu_0\|_{P,2} = o_p(1), \qquad \|\hat r^{(-k)} - r_0\|_{P,2} = o_p(1), \qquad \|\hat\pi^{(-k)} - \pi_0\|_{P,2} = o_p(1),$$

and

$$\|\hat\pi^{(-k)} - \pi_0\|_{P,2}\Big(\|\hat\mu^{(-k)} - \mu_0\|_{P,2} + |\tau_0|\,\|\hat r^{(-k)} - r_0\|_{P,2}\Big) = o_p(n^{-1/2}).$$

Then the estimator $\hat\tau$ from the nested-CV protocol in Section 2.4 satisfies

$$\sqrt{n}(\hat\tau - \tau_0) \xrightarrow{d} \mathcal{N}(0, V),$$

where V is given by (3.5). Moreover, the plug-in variance estimator satisfies

$$\hat V \to_p V,$$

so the Wald interval

$$\hat\tau \pm z_{1-\alpha/2}\sqrt{\hat V/n}$$

is asymptotically valid.

The asymptotic variance in Theorem 2 is

$$V = \big(g'_{ZD,0}\Omega_0^{-1}g_{ZD,0}\big)^{-1}\big(g'_{ZD,0}\Omega_0^{-1}\Sigma_0\,\Omega_0^{-1}g_{ZD,0}\big)\big(g'_{ZD,0}\Omega_0^{-1}g_{ZD,0}\big)^{-1}, \tag{3.5}$$

where

$$\Omega_0 := E[\tilde Z_i\tilde Z_i'], \qquad g_{ZD,0} := E[\tilde Z_i\tilde D_i], \qquad \Sigma_0 := E[\psi_i(\tau_0,\eta_0)\psi_i(\tau_0,\eta_0)'].$$

When $d_z = 1$, this reduces to the familiar scalar just-identified IV variance formula based on the residualized orthogonal score.

Proof. See Printed Appendix A.

Remark 3 (Strong versus weak instruments). The asymptotic normality result in Theorem 2 relies on the IV relevance condition in Assumption 2. In the vector-instrument case, this can be interpreted as requiring that the concentration quantity $g'_{ZD,0}\Omega_0^{-1}g_{ZD,0}$ is bounded away from zero (uniformly in n), so that the Jacobian of the score with respect to τ is well behaved and the asymptotic variance in (3.5) is finite. When instruments are weak (so $g'_{ZD,0}\Omega_0^{-1}g_{ZD,0}$ is small), classical strong-IV Gaussian approximations may be poor and alternative weak-instrument-robust infer-

ence procedures may be preferable. The focus of this paper is on leakage-safe nuisance selection for DML-IV under strong identification. Developing selector-specific weak-IV robust inference within this leakage-safe tuning protocol is beyond the scope of the paper.

Let $\tilde{Y}_i, \tilde{D}_i, \tilde{Z}_i$ denote the cross-fitted residuals produced by the nested-CV protocol in Section 2.4. Define

$$\hat{\Omega} = \frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i \tilde{Z}_i', \qquad \hat{g}_{ZD} = \frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i \tilde{D}_i, \qquad \hat{\psi}_i = \tilde{Z}_i(\tilde{Y}_i - \hat{\tau}\tilde{D}_i), \qquad \hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} \hat{\psi}_i \hat{\psi}_i'.$$

A consistent plug-in estimator of V in (3.5) is

$$\widehat{V} = \left(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}\right)^{-1}\left(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{\Sigma}\,\hat{\Omega}^{-1}\hat{g}_{ZD}\right)\left(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}\right)^{-1}.$$

I report $\widehat{SE}(\hat{\tau}) = \sqrt{\widehat{V}/n}$ and two-sided 95% confidence intervals $\hat{\tau} \pm 1.96\,\widehat{SE}(\hat{\tau})$.

3.6. Efficiency considerations

The estimator is designed for *regular* $\sqrt{n}$ inference under high-dimensional nuisance estimation. Full semiparametric efficiency in IV models requires an optimal instrument function and, under heteroskedasticity, inverse conditional variance weighting; see Chamberlain (1987) and Newey (1990). The baseline implementation uses the residualized instrument $\tilde{Z} = Z - \pi_0(W)$ and a 2SLS-style weight Ω_0^{-1}. This choice is convenient, robust, and well aligned with applied IV practice; it coincides with an efficient influence function only under additional restrictions (e.g. conditions under which the chosen instrument/weight matches the optimal instrument up to scale). Accordingly, the main claims in this paper are: (i) identification, (ii) Neyman orthogonality, (iii) $\sqrt{n}$-consistency and asymptotic normality with a consistent plug-in variance estimator, and (iv) finite-sample performance gains from disciplined nested-CV nuisance selection.

4. MONTE CARLO EVIDENCE

This section evaluates whether leakage-safe nested selection changes finite-sample performance relative to fixing a nuisance learner ex ante. I consider one sparse design, in which penalized linear learners are well aligned with the nuisance structure, and one nonlinear design, in which flexible learners are favored. Full data-generating equations, learner grids, and complete simulation tables are reported in the online appendix.

4.1. Design and implementation

In each design, the data follow the PLIV model with high-dimensional controls, a vector-valued instrument, and sample sizes $n \in \{1000, 3000, 10000\}$. Each simulated dataset is estimated using the leakage-safe nested-CV protocol with $K = 3$ outer folds and $J = 3$ inner folds. The searched nuisance library includes penalized linear learners, tree-based and boosting methods, and an MLP baseline; for the instrument regression $\pi_0(W) = E[Z \mid W]$ with $d_z > 1$, estimation is carried out coordinate-wise. The main-text summary reports RMSE and empirical coverage of nominal 95% confidence intervals. Full results for mean estimates, bias, standard deviations, estimated standard errors, all learner classes, and the $n = 10{,}000$ design are reported in the online appendix. The simulation designs are intended to evaluate fold-specific nuisance selection under standard strong-identification PLIV environments rather than to model identification-destructive residualization directly. In particular, the current designs keep the instrument residualization step relatively benign, so the Monte Carlo evidence should be read as validating the selector

in standard PLIV settings, while the CollegeDistance application and its appendix diagnostics provide the paper's main empirical illustration of residualized-first-stage fragility under flexible nuisance learning.

4.2. Key results

Table 1 shows two robust patterns. In the sparse design, Nested-CV is essentially tied with the strongest penalized-linear baseline: at $n = 1000$ the RMSE is 0.138 for both Nested-CV and Elastic Net, and by $n = 3000$ the difference is only 0.001. In the nonlinear design, Nested-CV improves materially on the linear baseline and performs comparably to a strong nonlinear fixed learner: relative to Elastic Net, RMSE falls from 0.307 to 0.270 at $n = 1000$ and from 0.144 to 0.123 at $n = 3000$, while coverage remains near nominal. The main message is therefore not universal dominance, but reduced sensitivity to arbitrary ex ante learner choice.

Table 1: Key Monte Carlo evidence

Design	Estimator	RMSE ($n = 1000$)	Cov. ($n = 1000$)	RMSE ($n = 3000$)	Cov. ($n = 3000$)
Sparse	Nested-CV	0.138	93	0.084	98
Sparse	Elastic Net	0.138	93	0.083	98
Sparse	Random Forest	0.148	95	0.090	97
Nonlinear	Nested-CV	0.270	90	0.123	98
Nonlinear	Elastic Net	0.307	92	0.144	98
Nonlinear	Random Forest	0.277	89	0.127	95

Notes: Entries report RMSE and empirical coverage (%) of nominal 95% confidence intervals for the cross-fitted PLIV estimator. Results are based on 100 replications with $K = 3$ outer folds and $J = 3$ inner folds. The sparse design favors penalized linear learners; the nonlinear design favors flexible learners. Full simulation details and complete results are reported in the online appendix.

5. EMPIRICAL APPLICATION: COLLEGEDISTANCE

This section illustrates the practical use of the proposed leakage-free Nested-CV DML-IV protocol in a standard textbook-style IV illustration using the `CollegeDistance` data Rouse (1995). The outcome is log wages, the endogenous regressor is years of education, and the excluded instrument is distance to a four-year college, controlling for a standard set of observed covariates. I report baseline OLS and homoskedastic TSLS estimates alongside DML-IV estimates based on cross-fitting with Nested-CV nuisance selection.

5.1. Implementation details and a sensitivity design for the instrument regression

The algorithm selects nuisance learners by nested cross-validation within each training fold. In IV settings, the instrument regression $\pi_0(W) = \mathbb{E}[Z \mid W]$ plays a distinct role: overly flexible learners for π_0 may absorb a large share of the identifying variation in Z, yielding residualized instruments with limited remaining variation. Accordingly, two Nested-CV specifications are reported: (i) a "full" π-library specification, which mirrors the Monte Carlo design, and (ii) a "restricted" π-library specification that limits π_0 to linear and regularized linear learners. The restricted π-library results are interpreted as an empirically motivated sensitivity design rather than as a replacement of the baseline procedure. Additional empirical robustness exercises are reported in Appendix E.

5.2. Results

Table 2 reports classical baselines TSLS and several DML-IV specifications. The purpose of this application is methodological: to illustrate how the proposed leakage-safe Nested-CV protocol disciplines nuisance selection and makes functional-form sensitivity transparent in a standard PLIV setting. Across both control specifications, the classical TSLS benchmark yields economically meaningful positive estimates (e.g., under W_{base}, $\hat{\tau} = 0.0519$ with SE $= 0.0200$), while OLS is close to zero. Fixed-learner DML-IV based on regularized linear nuisance fits produces estimates similar in magnitude to TSLS. In contrast, allowing highly flexible nuisance learners (e.g., Random Forest and boosting) yields estimates close to zero. This divergence underscores a practical vulnerability in applied DML-IV: absent a disciplined protocol, a researcher can obtain qualitatively different causal estimates across seemingly reasonable nuisance specifications.

Table 2: CollegeDistance: PLIV estimates of the return to schooling

	Baseline controls (W_{base}, $p = 11$)		High-dimensional controls (W_{hd}, $p = 77$)	
Estimator	$\hat{\tau}$	SE	$\hat{\tau}$	SE
2SLS (benchmark)[a]	0.0519	0.0200	0.0519	0.0200
DML-IV (Ridge)	0.0537	0.0197	0.0693	0.0226
DML-IV (Random Forest)	0.0015	0.0016	0.0026	0.0021
Nested-CV (full π library)	0.0016	0.0018	0.0029	0.0024
Nested-CV (restricted π library)[b]	-0.0003	0.0004	0.0013	0.0012

[a] The 2SLS benchmark uses the baseline controls W_{base} and is repeated in the high-dimensional column for ease of comparison.
[b] The restricted π-library limits the instrument regression $\pi_0(W) = \mathbb{E}[Z \mid W]$ to linear and regularized linear learners.
[c] The OLS benchmark and the full set of fixed-learner DML-IV specifications are reported in Appendix E.
[d] $n = 4{,}739$. Outcome is log wage; treatment is years of education; instrument is distance to a four-year college. DML-IV uses outer cross-fitting with $K = 5$ folds and leakage-free nested CV with $K_{\text{inner}} = 3$ folds within each training split. Standard errors are based on the orthogonal PLIV score.
[e] The very small standard errors reported for the zero-centered flexible-learner specifications are not a scaling artifact: throughout the paper, they are computed as $\widehat{SE}(\hat{\tau}) = \sqrt{\hat{V}/n}$ from the plug-in orthogonal-score variance formula in Section 3.5. Appendix E shows that these estimates are accompanied by modest residualized-first-stage diagnostics (Table 5) and tight repeat-split concentration around zero (Table 6), which is consistent with a setting in which the cross-fitted partialling-out step leaves limited residualized instrument variation under the chosen learner specifications.

The Nested-CV estimator resolves this researcher-choice problem by selecting nuisance learners based on strictly held-out inner-fold prediction loss within each outer cross-fitting split. When the learner library is unrestricted for all nuisance functions (full π library), Nested-CV selects highly flexible learners for $\pi_0(W) = E[Z \mid W]$ and the resulting DML-IV estimates are close to zero. To assess sensitivity to potential over-residualization of the excluded instrument, I also report a restricted-π specification in which $\pi_0(W)$ is selected only from linear and regularized linear learners, while $\mu_0(W)$ and $r_0(W)$ remain selected from the full library. Under this restriction, the Nested-CV estimates remain close to zero, indicating that the sensitivity in this dataset is not driven solely by the instrument regression, but also by the flexibility allowed in the outcome and treatment nuisance regressions. Note that this empirical section is not intended to adjudicate the substantive returns-to-schooling question. Rather, it demonstrates that the proposed leakage-safe Nested-CV protocol (i) enforces reproducible nuisance selection, (ii) prevents leakage in tuning/selection, and (iii) transparently documents the extent to which DML-IV estimates can be sensitive to nuisance-learning choices. Appendix E reports full model output, selection frequencies, residualized-instrument diagnostics, and repeat-split stability checks.

6. CONCLUSION

This paper studies leakage-safe fold-specific nuisance tuning for cross-fitted DML-IV in the partially linear IV model. The contribution is methodological and implementation-oriented: the standard orthogonal score and residualized 2SLS/GMM structure are retained, while nuisance learners and tuning parameters are selected within each outer training fold so that score evaluation remains separated from model selection. For finite searched libraries under squared-loss inner cross-validation, and under a balanced foldwise margin and conditional Bernstein-type fast-rate condition, the paper provides a selector-specific excess-risk bound showing that the selected refitted nuisance achieves benchmark-relative excess risk up to a $\log N/n$ term and a refit-stability remainder. Under the resulting high-level product-rate condition, the cross-fitted PLIV estimator is asymptotically normal with plug-in variance estimation. The paper does not benchmark against deliberately leaky tuning workflows such as full-sample cross-validation reused across folds, and it does not provide primitive sufficient conditions for every off-the-shelf learner in the empirical library. Its formal contribution is instead an assumption-driven finite-library rate-transfer result together with a leakage-safe implementation protocol, while the empirical sections illustrate the practical sensitivity that remains under modern nuisance-learning choices.

Monte Carlo experiments show that the proposed protocol remains competitive with the strongest fixed learner specifications in sparse designs and performs especially well in nonlinear designs. The CollegeDistance application illustrates substantial sensitivity to nuisance specification and the practical value of leakage-safe, reproducible tuning, while the Angrist–Krueger appendix provides an auxiliary stress test of the workflow under limited identifying variation, even though the formal asymptotic results of the paper are stated under strong identification.

There are several natural extensions. First, settings with potentially invalid instruments motivate combining DML-style orthogonality with selection and robustness to exclusion violations Sun, Cui, and Tchetgen (2022). Natural directions for future research also include sharper selector-specific excess-risk theory for the nested rule itself, learner aggregation or stacking within the same leakage-safe framework, and weak-identification-robust extensions of the procedure. The proposed estimator is complementary to automatic DML approaches that learn debiasing weights or optimal instruments directly Chen et al. (2021); Chernozhukov, Newey, and Singh (2022b,c); understanding how these approaches interact with leakage-safe nuisance selection is an interesting open question.

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A. APPENDIX: MAIN SELECTOR AND INFERENCE PROOFS

A.1. Proof of Lemma 4

Fix a target T and an outer fold k. Write $g^\dagger := g^\dagger_{T,n}$. For each candidate $g \in \mathcal{G}_{T,n}$, define

$$\widehat{\Gamma}_{T,k}(g) := \widehat{R}_{T,k}(g) - \widehat{R}_{T,k}(g^\dagger), \qquad \bar{\Gamma}_{T,k}(g) := \bar{R}_{T,k}(g) - \bar{R}_{T,k}(g^\dagger).$$

Because $\hat{g}_{T,k}$ minimizes $\widehat{R}_{T,k}(g)$ over $g \in \mathcal{G}_{T,n}$, one has

$$\widehat{\Gamma}_{T,k}(\hat{g}_{T,k}) \le 0.$$

For each $g \in \mathcal{G}_{T,n}$ and $m = 1, \dots, J$, define

$$A_{g,k,m} := \frac{1}{|\mathcal{I}_{-k,m}|} \sum_{i \in \mathcal{I}_{-k,m}} \Big(\Delta_{g,k,m}(O_i) - \gamma_{g,k,m} \Big),$$

so that

$$\widehat{\Gamma}_{T,k}(g) - \bar{\Gamma}_{T,k}(g) = \frac{1}{J} \sum_{m=1}^{J} A_{g,k,m}.$$

Conditional on $\mathcal{F}^{\mathrm{tr}}_{k,m}$, the validation observations $\{O_i : i \in \mathcal{I}_{-k,m}\}$ are independent of the fitted predictors $\hat{f}^{(-k,-m)}_g$ and $\hat{f}^{(-k,-m)}_{g^\dagger}$, since both are trained only on $\mathcal{I}_{-k} \setminus \mathcal{I}_{-k,m}$ and the benchmark index $g^\dagger$ is nonrandom. Hence the summands in $A_{g,k,m}$ are conditionally i.i.d. mean zero. By Assumption 6(ii)–(iii), Bernstein's inequality yields a constant $c_1 > 0$ such that for every $u > 0$,

$$P\big(A_{g,k,m} \le -u \,\big|\, \mathcal{F}^{\mathrm{tr}}_{k,m}\big) \le \exp\left(-c_1 |\mathcal{I}_{-k,m}| \min\left\{ \frac{u^2}{\gamma_{g,k,m}},\, u \right\} \right),$$

uniformly over g, m, k.

Now fix $g \in \mathcal{G}_{T,n}$. If $\widehat{\Gamma}_{T,k}(g) \le 0$, then

$$\frac{1}{J} \sum_{m=1}^{J} A_{g,k,m} \le -\frac{1}{J} \sum_{m=1}^{J} \gamma_{g,k,m}.$$

Therefore there must exist at least one inner fold m such that

$$A_{g,k,m} \le -\gamma_{g,k,m}.$$

Consequently, for any $t > 0$,

$$\{\widehat{\Gamma}_{T,k}(g) \le 0,\ \bar{\Gamma}_{T,k}(g) > t\} \subseteq \bigcup_{m=1}^{J} \{A_{g,k,m} \le -\gamma_{g,k,m},\ \gamma_{g,k,m} \ge \underline{c}\, t\},$$

where the second event uses Assumption 6(i).

On the event $\{\gamma_{g,k,m} \ge \underline{c}\, t\}$, the Bernstein bound with $u = \gamma_{g,k,m}$ gives

$$P\big(A_{g,k,m} \le -\gamma_{g,k,m} \,\big|\, \mathcal{F}^{\mathrm{tr}}_{k,m}\big) \le \exp(-c_2 |\mathcal{I}_{-k,m}|\, \gamma_{g,k,m}) \le \exp(-c_3 n t)$$

for constants $c_2, c_3 > 0$, uniformly over g, m, k, since $|\mathcal{I}_{-k,m}| \asymp n$ and $\gamma_{g,k,m} \ge \underline{c}\, t$.

Thus

$$\begin{aligned} P\Big(\widehat{\Gamma}_{T,k}(g) \le 0,\ \bar{\Gamma}_{T,k}(g) > t\Big) &\le \sum_{m=1}^{J} P(A_{g,k,m} \le -\gamma_{g,k,m},\ \gamma_{g,k,m} \ge \underline{c}\, t) \\ &\le J \exp(-c_3 n t). \end{aligned}$$

Choose

$$t = M\frac{\log N_{T,n}}{n}.$$

A union bound over $g \in \mathcal{G}_{T,n}$ then gives

$$P\left(\exists g \in \mathcal{G}_{T,n} : \widehat{\Gamma}_{T,k}(g) \le 0,\ \bar{\Gamma}_{T,k}(g) > M\frac{\log N_{T,n}}{n}\right) \le J N_{T,n}^{1-c_3 M}.$$

Choosing M sufficiently large implies that the right-hand side converges to zero uniformly in k. Since $\widehat{\Gamma}_{T,k}(\hat{g}_{T,k}) \le 0$, it follows that

$$[\bar{\Gamma}_{T,k}(\hat{g}_{T,k})]_+ = O_p\left(\frac{\log N_{T,n}}{n}\right),$$

uniformly in k.

Finally,

$$\begin{aligned} R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,k}^{\text{full}}(g_{T,n}^{\dagger}) &= \left(R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - \bar{R}_{T,k}(\hat{g}_{T,k})\right) + \bar{\Gamma}_{T,k}(\hat{g}_{T,k}) \\ &\quad + \left(\bar{R}_{T,k}(g_{T,n}^{\dagger}) - R_{T,k}^{\text{full}}(g_{T,n}^{\dagger})\right). \end{aligned}$$

Assumption 5(ii) implies that the two refit-gap terms are $O_p(s_{T,n})$, so

$$\left(R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,k}^{\text{full}}(g_{T,n}^{\dagger})\right)_+ = O_p\left(\frac{\log N_{T,n}}{n} + s_{T,n}\right),$$

uniformly in k.

A.2. Proof of Lemma 5 and Corollary 1

By Lemma 4,

$$\left(R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,k}^{\text{full}}(g_{T,n}^{\dagger})\right)_+ = O_p\left(\frac{\log N_{T,n}}{n} + s_{T,n}\right),$$

uniformly in k. By Assumption 5(i),

$$R_{T,k}^{\text{full}}(g_{T,n}^{\dagger}) - R_{T,0} = O_p(b_{T,n}^2),$$

uniformly in k. Therefore

$$R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,0} \le \left(R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,k}^{\text{full}}(g_{T,n}^{\dagger})\right)_+ + \left(R_{T,k}^{\text{full}}(g_{T,n}^{\dagger}) - R_{T,0}\right),$$

which implies

$$R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,0} = O_p\left(b_{T,n}^2 + \frac{\log N_{T,n}}{n} + s_{T,n}\right),$$

uniformly in k.

Under squared loss and $m_{T,0}(X) = E[T \mid X]$,

$$R_{T,k}^{\text{full}}(\hat{g}_{T,k}) - R_{T,0} = \|\hat{f}_{\hat{g}_{T,k}}^{(-k)} - m_{T,0}\|_{P,2}^2.$$

Taking square roots yields

$$\|\hat{f}_{\hat{g}_{T,k}}^{(-k)} - m_{T,0}\|_{P,2} = O_p\left(b_{T,n} + \sqrt{\frac{\log N_{T,n}}{n}} + \sqrt{s_{T,n}}\right),$$

uniformly in k.

For Corollary 1, apply this bound to $T = Y$, $T = D$, and each coordinate $T = Z_j$. Since d_z is fixed,

$$\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} = O_p\left(\max_{1\le j\le d_z} \rho_{\pi_j,n}\right) = O_p(\rho_{\pi,n}),$$

uniformly in k. The stated product-rate condition then implies that the second-order remainder in Lemma 3 is $o_p(n^{-1/2})$ uniformly in k. The conclusion follows from Theorem 2.

A.3. Auxiliary bound for the score difference

Lemma 6 (Score-difference second-moment bound). Suppose Assumption 3 holds. For outer fold k, define

$$\Delta^{(-k)}(W) := (\hat{\mu}^{(-k)} - \mu_0)(W) - \tau_0(\hat{r}^{(-k)} - r_0)(W)$$

and

$$\delta(O; \tau_0, \hat{\eta}^{(-k)}, \eta_0) := \psi(O; \tau_0, \hat{\eta}^{(-k)}) - \psi(O; \tau_0, \eta_0).$$

Then, uniformly in k,

$$E\Big[\|\delta(O; \tau_0, \hat{\eta}^{(-k)}, \eta_0)\|_2^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \lesssim O_p(1)\Big(\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2}^2 + \|\hat{\mu}^{(-k)} - \mu_0\|_{P,2}^2 + \|\hat{r}^{(-k)} - r_0\|_{P,2}^2\Big),$$

where $\mathcal{F}_k^{\mathrm{tr}} := \sigma\{O_j : j \in \mathcal{I}_{-k}\}$.

Proof. Let $\zeta := Y - \mu_0(W) - \tau_0(D - r_0(W))$ and $\tilde{Z}_0 := Z - \pi_0(W)$. Then

$$\delta(O; \tau_0, \hat{\eta}^{(-k)}, \eta_0) = -(\hat{\pi}^{(-k)} - \pi_0)\zeta - (Z - \hat{\pi}^{(-k)}(W))\,\Delta^{(-k)}(W).$$

Hence

$$\|\delta\|_2^2 \le 2\|\hat{\pi}^{(-k)} - \pi_0\|_2^2\zeta^2 + 2\|Z - \hat{\pi}^{(-k)}(W)\|_2^2\big(\Delta^{(-k)}(W)\big)^2.$$

Conditional on $\mathcal{F}_k^{\mathrm{tr}}$, the nuisance estimates are fixed functions of W. Using $E[\zeta^2 \mid W] \le C$,

$$E\Big[\|\hat{\pi}^{(-k)} - \pi_0\|_2^2\zeta^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \le C\,\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2}^2.$$

Next write

$$Z - \hat{\pi}^{(-k)}(W) = \tilde{Z}_0 + (\pi_0 - \hat{\pi}^{(-k)})(W),$$

so

$$\|Z - \hat{\pi}^{(-k)}(W)\|_2^2 \le 2\|\tilde{Z}_0\|_2^2 + 2\|\hat{\pi}^{(-k)} - \pi_0\|_2^2.$$

Therefore

$$\begin{aligned} E\Big[\|Z - \hat{\pi}^{(-k)}(W)\|_2^2\big(\Delta^{(-k)}(W)\big)^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] &\le 2E\Big[\|\tilde{Z}_0\|_2^2\big(\Delta^{(-k)}(W)\big)^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \\ &\quad + 2E\Big[\|\hat{\pi}^{(-k)} - \pi_0\|_2^2\big(\Delta^{(-k)}(W)\big)^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big]. \end{aligned}$$

Using $E[\|\tilde{Z}_0\|_2^2 \mid W] \le C$,

$$E\Big[\|\tilde{Z}_0\|_2^2\big(\Delta^{(-k)}(W)\big)^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \le C\,\|\Delta^{(-k)}\|_{P,2}^2.$$

Also, by the sup-norm boundedness in Assumption 3,

$$E\Big[\|\hat{\pi}^{(-k)} - \pi_0\|_2^2\big(\Delta^{(-k)}(W)\big)^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \le \|\hat{\pi}^{(-k)} - \pi_0\|_\infty^2\,\|\Delta^{(-k)}\|_{P,2}^2 = O_p(1)\,\|\Delta^{(-k)}\|_{P,2}^2.$$

Finally,

$$\|\Delta^{(-k)}\|_{P,2}^2 \le 2\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2}^2 + 2\tau_0^2\|\hat{r}^{(-k)} - r_0\|_{P,2}^2.$$

Combining the previous displays yields the claim.

A.4. Consistency of the plug-in variance estimator

Lemma 7 (Consistency of the plug-in variance estimator). Suppose the conditions of Theorem 2 and Assumption 3 hold. Then

$$\hat{\Omega} \to_p \Omega_0, \qquad \hat{g}_{ZD} \to_p g_{ZD,0}, \qquad \hat{\Sigma} \to_p \Sigma_0,$$

and consequently

$$\hat{V} \to_p V.$$

Proof. Write $\tilde{Z}_i = \tilde{Z}_{0,i} - (\hat{\pi}^{(-k(i))} - \pi_0)(W_i)$ and $\tilde{D}_i = \tilde{D}_{0,i} - (\hat{r}^{(-k(i))} - r_0)(W_i)$. Then

$$\hat{\Omega} - \mathbb{E}_n[\tilde{Z}_{0,i}\tilde{Z}'_{0,i}] = -2\mathbb{E}_n\left[\tilde{Z}_{0,i}(\hat{\pi}^{(-k(i))} - \pi_0)(W_i)'\right] + \mathbb{E}_n\left[(\hat{\pi}^{(-k(i))} - \pi_0)(W_i)(\hat{\pi}^{(-k(i))} - \pi_0)(W_i)'\right].$$

By Cauchy–Schwarz, the bounded conditional second moment of $\tilde{Z}_0$, and $\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2} = o_p(1)$, the right-hand side is $o_p(1)$. Since $\mathbb{E}_n[\tilde{Z}_{0,i}\tilde{Z}'_{0,i}] \to_p \Omega_0$ by the LLN, $\hat{\Omega} \to_p \Omega_0$.

Similarly,

$$\hat{g}_{ZD} - \mathbb{E}_n[\tilde{Z}_{0,i}\tilde{D}_{0,i}] = -\mathbb{E}_n\left[\tilde{Z}_{0,i}(\hat{r}^{(-k(i))} - r_0)(W_i)\right] - \mathbb{E}_n\left[(\hat{\pi}^{(-k(i))} - \pi_0)(W_i)\tilde{D}_{0,i}\right] + \mathbb{E}_n\left[(\hat{\pi}^{(-k(i))} - \pi_0)(W_i)(\hat{r}^{(-k(i))} - r_0)(W_i)\right],$$

and each term is $o_p(1)$ by Cauchy–Schwarz, the bounded conditional second moments of $\tilde{Z}_0$ and $\tilde{D}_0$, and $L_2(P)$-consistency of $\hat{r}^{(-k)}$ and $\hat{\pi}^{(-k)}$. Since $\mathbb{E}_n[\tilde{Z}_{0,i}\tilde{D}_{0,i}] \to_p g_{ZD,0}$, it follows that $\hat{g}_{ZD} \to_p g_{ZD,0}$.

Now let

$$\psi_{0,i} := \tilde{Z}_{0,i}\big(\tilde{Y}_{0,i} - \tau_0\tilde{D}_{0,i}\big)$$

and

$$\hat{\psi}_i := \tilde{Z}_i(\tilde{Y}_i - \hat{\tau}\tilde{D}_i).$$

Decompose

$$\hat{\psi}_i - \psi_{0,i} = \left[\psi(O_i; \tau_0, \hat{\eta}^{(-k(i))}) - \psi(O_i; \tau_0, \eta_0)\right] - (\hat{\tau} - \tau_0)\tilde{Z}_i\tilde{D}_i.$$

By Lemma 6 and $\hat{\tau} - \tau_0 = o_p(1)$, one has

$$\mathbb{E}_n\left[\|\hat{\psi}_i - \psi_{0,i}\|_2^2\right] = o_p(1).$$

Therefore

$$\hat{\Sigma} - \mathbb{E}_n[\psi_{0,i}\psi'_{0,i}] = \mathbb{E}_n\left[(\hat{\psi}_i - \psi_{0,i})\psi'_{0,i}\right] + \mathbb{E}_n\left[\psi_{0,i}(\hat{\psi}_i - \psi_{0,i})'\right] + \mathbb{E}_n\left[(\hat{\psi}_i - \psi_{0,i})(\hat{\psi}_i - \psi_{0,i})'\right] = o_p(1)$$

by Cauchy–Schwarz and the moment condition $E[\|\psi_{0,i}\|_2^{2+\delta}] < \infty$. Since $\mathbb{E}_n[\psi_{0,i}\psi'_{0,i}] \to_p \Sigma_0$, it follows that $\hat{\Sigma} \to_p \Sigma_0$.

Finally, Ω_0 is positive definite by Assumption 2, so $\hat{\Omega}$ is nonsingular with probability approaching one and the generalized inverse allowed by the estimator is asymptotically equivalent to the ordinary inverse. Continuous mapping then yields $\hat{V} \to_p V$.

A.5. Proof of Theorem 2 (Consistency and asymptotic normality)

I prove the result allowing for a fixed-dimensional vector instrument $Z \in \mathbb{R}^{d_z}$, with $d_z \geq 1$ fixed. Let $k(i)$ be the fold index such that $i \in \mathcal{I}_{k(i)}$, and recall the cross-fitted residuals

$$\tilde{Z}_i := Z_i - \hat{\pi}^{(-k(i))}(W_i), \qquad \tilde{Y}_i := Y_i - \hat{\mu}^{(-k(i))}(W_i), \qquad \tilde{D}_i := D_i - \hat{r}^{(-k(i))}(W_i).$$

Define the cross-fitted empirical moments

$$\hat{\Omega} := \mathbb{E}_n[\tilde{Z}_i\tilde{Z}_i'], \qquad \hat{g}_{ZD} := \mathbb{E}_n[\tilde{Z}_i\tilde{D}_i], \qquad \hat{g}_{ZY} := \mathbb{E}_n[\tilde{Z}_i\tilde{Y}_i].$$

The estimator defined by the nested-CV protocol in Section 2.4 is

$$\hat{\tau} = \frac{\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZY}}{\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}}.$$

Equivalently, defining the (vector) sample moment function

$$\hat{g}(\tau) := \mathbb{E}_n\Big[\tilde{Z}_i(\tilde{Y}_i - \tau\tilde{D}_i)\Big] = \hat{g}_{ZY} - \tau\hat{g}_{ZD} \in \mathbb{R}^{d_z},$$

I have $\hat{\tau}$ as the weighted residualized GMM/2SLS solution:

$$\hat{\tau} = \arg\min_{\tau\in\mathbb{R}} \ \hat{g}(\tau)'\hat{\Omega}^{-1}\hat{g}(\tau),$$

whose first-order condition is

$$\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}(\hat{\tau}) = 0.$$

Step 1: Since $\hat{g}(\tau) = \hat{g}(\tau_0) - (\tau - \tau_0)\hat{g}_{ZD}$, the first-order condition yields

$$0 = \hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}(\hat{\tau}) = \hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}(\tau_0) - (\hat{\tau} - \tau_0)\,\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}.$$

Thus I have the exact identity

$$\hat{\tau} - \tau_0 = \Big(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}\Big)^{-1}\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}(\tau_0). \tag{A.6}$$

Step 2: Write $\psi_{0,i} := \psi(O_i;\tau_0,\eta_0) = \tilde{Z}_{0,i}(\tilde{Y}_{0,i} - \tau_0\tilde{D}_{0,i})$, where $\tilde{Z}_{0,i} = Z_i - \pi_0(W_i)$, $\tilde{Y}_{0,i} = Y_i - \mu_0(W_i)$, and $\tilde{D}_{0,i} = D_i - r_0(W_i)$. Then $E[\psi_{0,i}] = 0$ and $\Sigma_0 := E[\psi_{0,i}\psi_{0,i}']$ is finite under the standing moment conditions.

Also note that

$$\hat{g}(\tau_0) = \mathbb{E}_n\Big[\psi(O_i;\tau_0,\hat{\eta}^{(-k(i))})\Big].$$

For each outer fold k, define the foldwise training sigma-field

$$\mathcal{F}_k^{\mathrm{tr}} := \sigma\{O_j : j \in \mathcal{I}_{-k}\},$$

and write

$$R_{n,2,k} := \sqrt{n}\,\frac{|\mathcal{I}_k|}{n}E\Big[\psi(O;\tau_0,\hat{\eta}^{(-k)}) - \psi(O;\tau_0,\eta_0)\,\Big|\,\mathcal{F}_k^{\mathrm{tr}}\Big].$$

Then

$$\sqrt{n}\,\hat{g}(\tau_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{0,i} + R_{n,1} + R_{n,2}, \qquad R_{n,2} := \sum_{k=1}^{K}R_{n,2,k},$$

where $R_{n,1}$ is the centered empirical-process part treated below.

Bias term. For each fold k, conditional on the training sample $\mathcal{I}_{-k}$, Lemma 3 implies

$$\Big\|E[\psi(\tau_0,\hat{\eta}^{(-k)}) - \psi(\tau_0,\eta_0)\mid\mathcal{I}_{-k}]\Big\|_2 \lesssim \|\hat{\pi}^{(-k)} - \pi_0\|_{P,2}\Big(\|\hat{\mu}^{(-k)} - \mu_0\|_{P,2} + |\tau_0|\,\|\hat{r}^{(-k)} - r_0\|_{P,2}\Big).$$

Under the product-rate condition stated in Theorem 2 (uniformly in k), the right-hand side is $o_p(n^{-1/2})$, hence $\|R_{n,2}\|_2 = o_p(1)$.

Empirical process term. For each outer fold k, define $\mathrm{R}_{n,1,k} := \frac{1}{\sqrt{n}}\sum_{i\in\mathcal{I}_k}\bar{\delta}_i^{(-k)}, \qquad \delta_i^{(-k)} :=$

$\psi(O_i;\tau_0,\hat{\eta}^{(-k)}) - \psi(O_i;\tau_0,\eta_0)$,

$$\bar{\delta}_i^{(-k)} := \delta_i^{(-k)} - E[\delta_i^{(-k)} \mid \mathcal{F}_k^{\mathrm{tr}}], \qquad \mathcal{F}_k^{\mathrm{tr}} := \sigma\{O_j : j \in \mathcal{I}_{-k}\}.$$

Then

$$R_{n,1} = \sum_{k=1}^{K} R_{n,1,k}.$$

Conditional on $\mathcal{F}_k^{\mathrm{tr}}$, the variables $\{\bar{\delta}_i^{(-k)} : i \in \mathcal{I}_k\}$ are i.i.d. mean zero. Therefore

$$E\big[\|R_{n,1,k}\|_2^2 \,\big|\, \mathcal{F}_k^{\mathrm{tr}}\big] = \frac{1}{n}\sum_{i\in\mathcal{I}_k} E\Big[\|\bar{\delta}_i^{(-k)}\|_2^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big] \le \frac{|\mathcal{I}_k|}{n} E\Big[\|\delta(O;\tau_0,\hat{\eta}^{(-k)},\eta_0)\|_2^2 \,\Big|\, \mathcal{F}_k^{\mathrm{tr}}\Big].$$

By Lemma 6,

$$E\big[\|R_{n,1,k}\|_2^2 \,\big|\, \mathcal{F}_k^{\mathrm{tr}}\big] \lesssim \frac{|\mathcal{I}_k|}{n} O_p(1)\Big(\|\hat{\pi}^{(-k)} - \pi_0\|_{P,2}^2 + \|\hat{\mu}^{(-k)} - \mu_0\|_{P,2}^2 + \|\hat{r}^{(-k)} - r_0\|_{P,2}^2\Big) = o_p(1),$$

uniformly in k. Markov's inequality therefore gives $\|R_{n,1,k}\|_2 = o_p(1)$ for each k, and since K is fixed,

$$\|R_{n,1}\|_2 \le \sum_{k=1}^{K} \|R_{n,1,k}\|_2 = o_p(1).$$

Step 3: Using the same cross-fitting/LLN arguments (and the fact that d_z is fixed), I have

$$\hat{\Omega} \xrightarrow{p} \Omega_0, \qquad \hat{g}_{ZD} \xrightarrow{p} g_{ZD,0} := E[\tilde{Z}\tilde{D}].$$

By Assumption 2, Ω_0 is positive definite and $g_{ZD,0} \neq 0$, so

$$g_{ZD,0}'\Omega_0^{-1} g_{ZD,0} > 0.$$

Hence, by continuous mapping,

$$\Big(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}\Big)^{-1}\hat{g}_{ZD}'\hat{\Omega}^{-1} \xrightarrow{p} \big(g_{ZD,0}'\Omega_0^{-1}g_{ZD,0}\big)^{-1} g_{ZD,0}'\Omega_0^{-1}.$$

Step 4: Multiplying (A.6) by $\sqrt{n}$ and using Steps 2–3 gives

$$\sqrt{n}(\hat{\tau} - \tau_0) = \Big(\hat{g}_{ZD}'\hat{\Omega}^{-1}\hat{g}_{ZD}\Big)^{-1}\hat{g}_{ZD}'\hat{\Omega}^{-1}\sqrt{n}\,\hat{g}(\tau_0) \xrightarrow{d} \mathcal{N}(0,V),$$

where

$$V = \big(g_{ZD,0}'\Omega_0^{-1}g_{ZD,0}\big)^{-1}\big(g_{ZD,0}'\Omega_0^{-1}\Sigma_0\,\Omega_0^{-1}g_{ZD,0}\big)\big(g_{ZD,0}'\Omega_0^{-1}g_{ZD,0}\big)^{-1}.$$

This is exactly (3.5). The scalar-instrument case $d_z = 1$ follows as a special case.

Step 5: Lemma 7 gives

$$\hat{\Omega} \to_p \Omega_0, \qquad \hat{g}_{ZD} \to_p g_{ZD,0}, \qquad \hat{\Sigma} \to_p \Sigma_0,$$

and hence $\hat{V} \to_p V$. This establishes the asymptotic validity of the Wald interval based on $\hat{V}$.